

Certified Local Obstructions and Exact Wall-Stratified Continuation toward Hilbert’s Sixteenth Problem

Consolidated evidence from the `math1` and `math2` campaigns

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Abstract

This paper consolidates two bounded research campaigns toward Hilbert’s sixteenth problem, Part II. The general problem remains open: no value of $H(n)$, $H(2)$, $H(6)$, no uniform finiteness theorem, no complete configuration theorem, and no 20-cycle degree-six candidate is claimed. The retained contribution is narrower: exact conventions for the problem, audited reasons why pointwise Dulac finiteness does not supply a uniform degree-wise bound, a finite wall-stratified continuation certificate for one Llibre–Ramirez–Sadovskaia-style degree-six local atlas cell near $\rho = 200$, and local obstructions in the quadratic (H_{13}^3) successor analysis, including Δ_G strata and a one-way $Y = 0$ crossing obstruction.

1 Evidence classes and scope

Every formal theorem or proposition below is tagged by one of the five allowed evidence classes:

1. **Class 1: established cited result;**
2. **Class 2: proved here;**
3. **Class 3: exact finite certificate;**
4. **Class 4: conditional reduction;**
5. **Class 5: open gap or failed route.**

The tags are part of the statements. Computational certificates are used only over the finite algebraic objects and parameter boxes explicitly named in the statement. A successful program exit code is treated as verifier evidence, not as a mathematical proof by itself; the proof sketches spell out the exact finite object, arithmetic domain, sign or root-count argument, and coverage claim. Anything not bridged in this way is explicitly downgraded to evidence or an open gap.

2 Problem conventions

Let

$$X = (P, Q), \quad \dot{x} = P(x, y), \quad \dot{y} = Q(x, y),$$

where $P, Q \in \mathbb{R}[x, y]$, and put $\deg X = \max(\deg P, \deg Q)$, with the zero field assigned degree $-\infty$. For $n \geq 0$,

$$\mathcal{V}_{\leq n} = \{X : \deg X \leq n\}.$$

A periodic orbit is the image in $\mathbb{R}^2$ of a non-stationary periodic solution. It is counted geometrically: reparametrization, orientation, period, multiplicity of a Poincare fixed point, and cyclicity do not create additional elements. A limit cycle is a periodic-orbit image isolated among periodic-orbit images. Define

$$\text{LC}(X) = \{\text{geometric finite-plane limit cycles of } X\},$$

and

$$L(X) = \sup\{|A| : A \subseteq \text{LC}(X), A \text{ finite}\} \in \mathbb{N}_0 \cup \{\infty\}.$$

The Hilbert number used here is the extended supremum

$$H(n) = \sup_{X \in \mathcal{V}_{\leq n}} L(X).$$

Thus $H(n) = \infty$ means unbounded finite counts across the degree- $\leq n$ family, not necessarily one vector field with infinitely many cycles. If $H(n) = m < \infty$, then the maximum is attained, since all realized finite values are integers. The line at infinity in the Poincare compactification is an invariant boundary used in cyclicity arguments, but it is not an affine limit cycle and contributes no element to $\text{LC}(X)$.

Theorem 1 (Conventions and finite configuration invariant; **Class 2: proved here**, with **Class 1: established cited result** background for Dulac finiteness). *For finite affine cycle families, ambient configuration equivalence is exactly unlabelled rooted nesting-forest equivalence. Singularities, separatrices, stability, multiplicity, and the line at infinity are not nodes of the base cycle configuration. Moreover $H_=(n) = H(n)$ for the exact-degree supremum $H_=(n) = \sup_{\deg X=n} L(X)$, but this equality of suprema is not a complete phase-portrait reduction for each field.*

Proof. Distinct periodic orbits of a polynomial ODE are disjoint by uniqueness, and each periodic orbit is a Jordan curve by least-period injectivity. For finitely many disjoint Jordan curves, the relation $\gamma \prec \delta$ if γ lies in the bounded component of $\mathbb{R}^2 \setminus \delta$ has a unique parent for every non-root curve; hence its Hasse diagram is a rooted forest. A plane homeomorphism preserves bounded complementary components, so it preserves the forest. Conversely, a forest isomorphism identifies the boundary components of the outer sphere-with-holes region and each disk-with-holes region; the Jordan–Schoenflies theorem and cross-cuts extend those boundary maps and glue them into an ambient homeomorphism fixing infinity.

For $H_=(n) = H(n)$, take a finite subfamily $A \subseteq \text{LC}(X)$ of a degree- $d < n$ nonzero field X . Choose a line $x = a$ disjoint from the compact union of curves in A and set $Y = (x - a)^{n-d}X$. Near every curve of A , the multiplier is nonzero, so trajectories and isolation are preserved up to time change. Therefore every finite cycle count visible below degree n is visible at exact degree n ; the reverse inequality is inclusion. This proves equality of suprema, while deliberately not asserting preservation of all singularities or all nonselected global phase-portrait data. $\square$

Proposition 1 (Dulac finiteness is not a uniform Hilbert bound; **Class 2: proved here**, with **Class 1: established cited result** background). *The per-vector-field Dulac–Ilyashenko–Eccles theorem supplies*

$$\forall X \in \mathcal{V}_{\leq n} \exists M_X \quad L(X) \leq M_X,$$

not the uniform statement

$$\exists M_n \forall X \in \mathcal{V}_{\leq n} \quad L(X) \leq M_n.$$

Consequently, citing pointwise finiteness alone does not prove $H(n) < \infty$.

Proof. The first formula has the quantifier order $\forall X \exists M_X$; the second has $\exists M_n \forall X$. The latter implies the former, but the reverse implication is invalid without an additional compactness, stratification, valency, or cyclicity theorem that controls the constants uniformly over the degree- $\leq n$ parameter space. This logical gap is exactly the remaining uniform Hilbert-number problem. $\square$

3 Route A: exact LRS-style wall-stratified continuation

The `math1` campaign investigated Llibre–Ramirez–Sadovskaia-style Darboux/invariant-curve mechanisms in degree six. The relevant local atlas fixes $n = 6$, the full-triangle span-20 tail-size-two support, target $T = \{(1, 3), (2, 3)\}$, and selected tail $(1, 3)$ for the first row in a far-negative B -chamber. The project tracks real branches of algebraic curves by event walls: nonzero constant and leading coefficients, discriminant walls for u -root collisions, common-resultant walls with the comparison polynomial C , and critical-value walls for levels 0 and 1. Descartes and Sturm isolation supply one-dimensional root counts on the $\rho = 200$ slice; rational interpolation and Bernstein sign certificates supply the local two-parameter tube.

Proposition 2 (The $\rho = 200$ LRS local tube certificate; **Class 3: exact finite certificate**). *In the Route A first-row selected-tail model just described, with $\rho = -\lambda - 2$, $B = 500 + s$, $s \in (0, \infty)$, and*

$$\rho \in \left[\frac{944473296573929042739199}{4722366482869645213696}, \frac{944473296573929042739201}{4722366482869645213696} \right],$$

the finite certificate proves the following and only the following:

1. *the $\rho = 200$ slice is transported through all positive- s event walls in the listed first-row selected-tail atlas;*
2. *the split-resumed local two-parameter tube has 24 ordered event clusters, 25 bad-event-free s -cells, and no regular-level $E = 11$ witness in that finite model;*
3. *every local cell has at most 10 real regular-level curve roots in the counted side word;*
4. *the certificate does not prove the full $B \geq 500$ chamber, selected-row retirement, tail-size-two retirement, global $E = 11$ nonexistence, $H(6)$, CONFIG6, or uniform Hilbert finiteness.*

Proof. The finite algebraic object is the list of event polynomials recorded by the verifier:

event	$\deg_q$	$\deg_s$
R constant coefficient	0	4
R leading coefficient	4	0
R discriminant	60	54
R, C discriminant common root	16	16
R, C leading common root	12	12
critical value 0	32	32
critical value 1	96	96

where $q = \rho - 200$ is restricted to the rational interval of half-width $1/4722366482869645213696$. On the central slice $q = 0$, the verifier isolates all positive s -roots of these families using exact integer/rational Sturm and Descartes computations: the respective positive root counts are 0, 0, 3, 3, 2, 4, 12, giving 24 ordered clusters. Between adjacent clusters, signs of the event polynomials are fixed; therefore the real branch count word is locally constant.

For the two discriminant clusters where the naive graph bracket fails, the certificate uses weighted local coordinates and exact positivity isolation to show there is no real punctured branch in the weighted subbox. The adjacent side words agree in maximum count, so these events are real collisions with no count increase. For the two-parameter tube, 129 rational q -nodes $(-64, \dots, 64)$ are interpolated exactly; Bernstein sign certificates prove the same event signs on every cell and on the $s \geq 4096$ tail. The output line records `two_parameter_tube_transport_certified=True`, `regular_level_E11_candidate_found=False`, and all nonclaims listed above. The boundary-exit q -collar output records a predecessor audit of this tube but no separate terminal theorem is used here. $\square$

Proposition 3 (A selected successor wall certificate; **Class 3: exact finite certificate**). *For the separate Route A selected row $T = \{(2, 2), (3, 1)\}$, selected tail $(2, 2)$, the certified crossing of*

$$\text{pos:event263:I1:pair-resultant:(4,9):bottom}$$

after the event263-top/event191-bottom predecessor is harmless in the finite atlas sense: event263 retires on that branch, event554 remains active, $\text{max_projected_real_u_roots_bound} = 9$, $\text{max_curve_point_upper} = 18$, and no certified 20-point regular-level candidate is isolated. This is row-local selected-tail evidence only.

Proof. The verifier constructs a rational successor rectangle and checks the wall crossing by exact signs at the bottom edge, exact resultants in λ for the common- α numerator and denominator, and a Descartes deletion certificate with maximum excess zero. The union audit then covers the successor pieces with Bernstein boxes and negative sign certificates. The validation output confirms the result line and preserves the next blocker as event554. Since the proof object is one selected wall in one selected atlas branch, it cannot be promoted to an $H(6)$, CONFIG6, row-retirement, or global nonexistence theorem. $\square$

4 Route B: compactified quadratic transition analysis

The `math2` campaign worked in the compactified Dumortier–Roussarie–Rousseau quadratic program. The retained local model for (H_{13}^3) uses the RR weighted blow-up with invariant leaves $r\rho = \nu$, Type-II endpoint Dulac maps at P_1, P_2 , and a rescaled $D_{\bar{\mu}}$ field. The sidecar sources identify the direct nonboundary P_1 -to- P_2 successor as the first concrete unresolved successor after the boundary hemicycle contrast.

Proposition 4 (H313 direct Type-II composition remains an open packet; **Class 4: conditional reduction** and **Class 5: open gap or failed route**). *For $0 < B < 1/2$ in the RR (H_{13}^3) range, the nonboundary direct P_1 -to- P_2 successor reduces to a scalar endpoint comparison on the same leaf $r\rho = \nu$,*

$$V_A(r, \rho) = D_1^r(r, \rho) - S_A(D_2^r(U_A r, U_A^{-1} \rho), \nu),$$

with $U_A(0, 0) \neq 0$. After the connection constant term is imposed,

$$V_A = r^{\sigma_0} (C_{H313}^{(0)}(A, \nu) + C_{H313}^{(1)}(r, \rho, A)).$$

The checked RR derivation-division theorem does not apply directly because the two endpoint Type-II terms have the same base monomial on the actual same- r comparison. Therefore this packet is a conditional reduction/open gap, not a finite-cyclicity theorem.

Proof. The finite symbolic check records $a = 2/3$, sample $B = 1/3$, $\sigma = 1/2$, endpoint eigenvalue signs consistent with Type-II passages at both P_1 and P_2 , and `actual_same_r_thderdiv_delta` = 0. RR's theorem requires nonresonance of ratios of leading monomials. The direct comparison places both endpoint leading terms in the same r^{σ_0} scale, so the named theorem's hypothesis is absent. No alternate checked theorem supplies the missing finite-zero statement. $\square$

Proposition 5 ($\Delta_G > 0$ middle-saddle one-way crossing obstruction; **Class 2: proved here**). *Consider the RR H313 off-line Lienard reduction with $1/2 < a < 1$, $\bar{\mu}_1 \neq 0$, and $\Delta_G > 0$. Let $\xi_1 < \xi_2 < \xi_3$ be the three simple roots of*

$$G(X) = (X + \bar{\mu}_3)(aX^2 + \bar{\mu}_2) - \bar{\mu}_1,$$

and let $S_2 = (\xi_2, Y_s)$, $Y_s = -a\xi_2^2 - \bar{\mu}_2$, be the middle finite saddle. Then S_2 cannot be the middle vertex of an endpoint-compatible H313 chain $P_1 \rightarrow S_2 \rightarrow P_2$.

Proof. The root ordering and $\Delta_G > 0$ hypothesis give a unique middle saddle: $G'(\xi_2) < 0$, with hyperbolicity ratio

$$\rho = \frac{\sqrt{\tau_s^2 - 4G'(\xi_2)} - \tau_s}{\sqrt{\tau_s^2 - 4G'(\xi_2)} + \tau_s}, \quad \tau_s = (2a + 1)\xi_2 + \bar{\mu}_3.$$

The H313 endpoint branch has $Y = c/X + O(X^{-2})$, $c = -\bar{\mu}_1/(a + 1)$. Hence the sign of Y near P_1 is $\text{sign}(\bar{\mu}_1)$, while the sign near P_2 is $-\text{sign}(\bar{\mu}_1)$. On the line $Y = 0$, the original vector field satisfies $\dot{Y} = \bar{\mu}_1$, so crossings of $Y = 0$ are one-way. Therefore a $P_1 \rightarrow S_2$ H313 half requires $\text{sign}(Y_s) = \text{sign}(\bar{\mu}_1)$, whereas an $S_2 \rightarrow P_2$ H313 half requires $\text{sign}(Y_s) = -\text{sign}(\bar{\mu}_1)$. In the off-line bucket $Y_s \neq 0$, because $Y_s = 0$ at a finite singularity would force $G(\xi_2) = -\bar{\mu}_1$, contradicting $\bar{\mu}_1 \neq 0$. The two necessary sign conditions are incompatible. $\square$

Proposition 6 ($\Delta_G = 0$ semihyperbolic landing split; **Class 2: proved here**). *In the same RR H313 off-line model, a double-root stratum with $q/\tau > 0$ has the local semihyperbolic left-in/right-out orientation, but no off-line member $\bar{\mu}_1 \neq 0$ has both H313 endpoint landings. The only two-ended survivor in this packet is the invariant-line case $\bar{\mu}_1 = \bar{\mu}_2 = s = 0$, $Y = 0$, oriented $P_1 \rightarrow S_0 \rightarrow P_2$; its endpoint-matched Dulac/anchor packet remains open.*

Proof. For a double root s ,

$$\bar{\mu}_2 = -a(3s^2 + 2\bar{\mu}_3s), \quad \bar{\mu}_1 = -2as(s + \bar{\mu}_3)^2,$$

and, after translation, the normal form is

$$\dot{z} = u, \quad \dot{u} = \tau u + (2a + 1)zu - qz^2 - az^3,$$

where $\tau = (2a + 1)s + \bar{\mu}_3$ and $q = a(3s + \bar{\mu}_3)$. The case $q/\tau > 0$ is the local left-in/right-out saddle-node orientation. At the double root,

$$Y_s = 2as(s + \bar{\mu}_3), \quad \bar{\mu}_1 Y_s = -4a^2 s^2 (s + \bar{\mu}_3)^3.$$

The H313 endpoint signs and one-way $Y = 0$ crossing rule are the same as in Proposition 5. If $s + \bar{\mu}_3 > 0$, only the $S \rightarrow P_2$ half is sign-compatible; if $s + \bar{\mu}_3 < 0$, only the $P_1 \rightarrow S$ half is sign-compatible. Thus no off-line $\bar{\mu}_1 \neq 0$ double-root member gives both endpoint landings. The invariant-line case has $Y = 0$, $\dot{X} = aX^2$, and $Y/X^2 \rightarrow 0$ at both endpoints, so it remains an incidence candidate, but no finite-cyclicity theorem for its full endpoint-matched transition packet is supplied. $\square$

5 Synthesis

The two routes share a strategy. First compactify the phase portrait or parameterized algebraic family. Then stratify the parameter space by exact event walls: discriminants, resultants, endpoint signs, transition domains, and invariant leaves. On each stratum, replace qualitative continuation by exact sign, root-count, or transition data. Finally, certify local exclusions or identify the exact theorem missing for global gluing.

The common obstruction is global uniformity. Route A shows that an exact wall-stratified atlas can transport a chosen local model through many events without finding an $E = 11$ or 20-point candidate, but the certificate is explicitly finite and selected. Route B shows that several plausible H313 successor chains fail by endpoint or one-way crossing signs, while the surviving direct or semihyperbolic packets still require endpoint-matched Dulac/transition finite-zero theorems. Neither route covers all vector fields of a fixed degree.

6 Claim table and proof dependencies

Claim	Class	Main dependencies	Limit
Theorem 1	Class 2; Class 1 for Dulac background	math2/research/scope/ foundational_conventions_ and_configuration_ theorem.md; math1/research/SCOPE.md	Definitions and finite configurations only; no $H(n)$ bound.
Proposition 1	Class 2; Class 1 for Dulac background	same scope files; Ilyashenko/Ecalle citations	Logical quantifier audit only.
Proposition 2	Class 3	math1/computations/lrs_ n6_full_triangle_span20_ tail_size_two_largeB_ first_row_rho200_ interior_rdisc_slice_ exact_run.out; math1/computations/lrs_ n6_full_triangle_span20_ tail_size_two_largeB_ first_row_rho200_two_ parameter_tube_resume_ exact_run_v26.out	Local first-row selected-tail tube; not $H(6)$ or CONFIG6.
Proposition 3	Class 3	math1/CHECKPOINT.md; math1/computations/row22_ 31_event263_I1_pair_ resultant_4_9_bottom_ after_event263_top_exact_ run.out	One selected successor wall only.

Proposition 4	Class 4/5	<code>math2/computations/fq3081_h313_nonboundary_atlas.sidecar.json; math2/computations/fq3082_h313_direct_typeii_composition.sidecar.json</code>	Reduction and theorem-gap statement, not cyclicity.
Proposition 5	Class 2	<code>math2/computations/fq3221_offline_positive_discriminant_splitting_sidecar.json</code>	Only $\Delta_G > 0$ middle-saddle H313 chain.
Proposition 6	Class 2	<code>math2/computations/fq3223_h313_saddlenode_global_landing_sidecar.json; math2/computations/fq3222_residual_relevance_sidecar.json</code>	Landing split only; surviving saddle-node packet open.

The machine-readable dependency graph is in `PROOF_DEPENDENCIES.json`. The audit rows in `CLAIMS_AUDIT.json` are intended to be checked claim-by-claim against the theorem and proposition labels above.

7 Research process and route decisions

Route A began with lower-bound and fixed-degree realizability mechanisms, including Harnack/tropical and LRS sparse-curve constructions. Many degree-six search classes were retired by exact genus, patchworking, Reeb-graph, and root-budget obstructions. The surviving evidence most useful for this paper is not a global construction but the exact continuation discipline: list event walls, isolate their roots, certify successor behavior, and mark every failed candidate with its finite scope.

Route B began with the DRR compactified quadratic program. Source audits kept the RR boundary theorem for named boundary limit-periodic sets separated from full finite cyclicity of all successors. The H313 direct Type-II packet remains open because the available derivation-division theorem does not match the same-monomial endpoint composition. The $\Delta_G > 0$ and $\Delta_G = 0$ strata nevertheless yield useful obstructions: one-way $Y = 0$ crossing blocks two-sided H313 landing through the middle saddle or off-line saddle-node, while the invariant-line survivor is narrowed to an explicit remaining packet.

Adversarial checks changed the frontier by demoting several plausible proof paths: boundary cyclicity was not treated as full H313 cyclicity; same-coordinate and central-atlas data not present in the source were not invented; Route A's local rho tube was not promoted to a full chamber or Hilbert-number statement; and a missing boundary-exit terminal result was not used as a theorem.

8 Limitations and open problems

The following are not proved here:

1. Hilbert's sixteenth problem, Part II, in general;
2. uniform finiteness $H(n) < \infty$ for any new n ;

3. a value of $H(2)$, $H(6)$, or any $H(n)$;
4. a complete degree-six configuration theorem or CONFIG6;
5. a certified degree-six 20-cycle candidate;
6. full finite cyclicity of all H313 nonboundary successors;
7. a global gluing theorem turning the local Route A or Route B exclusions into degree-wise uniform bounds.

The remaining mathematical gap is not a lack of local signs; it is the absence of a theorem that glues all compactified strata, degenerate exact-degree faces, transition maps, and local finite-zero packets uniformly over the relevant coefficient space.

A Reproducibility appendix

The exact verifier entry points and expected outputs are as follows.

Scope	Entry point	Expected output
Route A $\rho = 200$ slice	<code>math1/computations/lrs_n6_full_triangle_span20_tail_size_two_largeB_first_row_rho200_interior_rdisc_slice_stratifier.py--certify</code>	<code>first_row_rho200_interior_rdisc_slice_st</code> no $E = 11$ witness.
Route A local tube	<code>math1/computations/lrs_n6_full_triangle_span20_tail_size_two_largeB_first_row_rho200_two_parameter_tube_verifier.py--resume-after-event-strips</code>	<code>first_row_rho200_two_parameter_tube_after</code> <code>two_parameter_tube_transport_certified=</code>
Route A boundary audit	<code>math1/computations/lrs_n6_full_triangle_span20_tail_size_two_largeB_first_row_rho200_boundary_verifier.py--certify</code>	predecessor audit only in this paper; no independent boundary theorem imported.
Route A successor	<code>math1/computations/row22_31_event263_I1_pair_resultant_4_9_bottom_after_event263_top.py</code>	harmless successor; ≤ 9 projected real u -roots, ≤ 18 curve points, no 20-point candidate.
Route B direct H313 packet	<code>math2/computations/fq3082_h313_direct_typeii_composition.sidecar.json</code>	theorem-gap decision: RR <code>thderiv</code> not applicable to same-monomial comparison.
Route B $\Delta_G > 0$	<code>math2/computations/fq3221_offline_positive_discriminant_splitting_validator.py</code>	validation passed; no endpoint-compatible $P_1 \rightarrow S_2 \rightarrow P_2$ H313 chain.

Route B residual relevance	math2/computations/fq3222_residual_relevance_validator.py	validation passed; no new residual $\Delta_G > 0$ annular/lower witness.
Route B $\Delta_G = 0$	math2/computations/fq3223_h313_saddlenode_global_landing_validator.py	validation passed; off-line semihyperbolic candidates lack both H313 landings; invariant-line survivor remains open.

All exact arithmetic cited above is rational/integer arithmetic. Root counts are Sturm or Descartes counts on explicitly isolated rational intervals. Sign coverage in two variables uses rational interpolation and Bernstein sign certificates on rational boxes. The finite domains and nonclaims are part of the verifier output and are repeated in CLAIMS_AUDIT.json.

B SHA-256 manifest of claim-bearing source files

Path	Bytes	SHA-256
math2/research/scope/foundational_conventions_and_configuration_theorem.md	13398	5aca76060de153e222ad2cb407ea282cac2b04e3720b67dbe7
math2/CONVENTIONS.md	11431	cf295f1d4a97ce616cf0bcf55da17f23470a5a2d49cc31e42
math1/research/SCOPE.md	26685	31b5ed1a92a23d8237d98da292c9c13033651201c62727f06f
math1/research/FIXED_DEGREE_REALIZATION.md	2774710	af7db9751697ac54944f4f51c64e433ec86d5e22f740fa41e1
math1/CHECKPOINT.md	2523	deae81fe3b94866b030843cfa4fbd4e1fc6be605ecd884245f
math1/computations/lrs_n6_full_triangle_span20_tail_size_two_largeB_first_row_rho200_interior_rdisc_slice_stratifier.py	51847	25f95b86d1f5a35ae35a5f4a4032cd4746b8961c24715996b5
math1/computations/lrs_n6_full_triangle_span20_tail_size_two_largeB_first_row_rho200_interior_rdisc_slice_exact_run.out	206438	ea65bce1e798b7bbd7c27ab4b14e3b308d4c5ffe247d66833
math1/computations/lrs_n6_full_triangle_span20_tail_size_two_largeB_first_row_rho200_two_parameter_tube_verifier.py	82833	b02f80cd6f716f3d788c19c55d30b851246eb1d3531938e3a
math1/computations/lrs_n6_full_triangle_span20_tail_size_two_largeB_first_row_rho200_two_parameter_tube_exact_run_v14.out	35506	78eb124a38925872adbcc350612ce21d3359ea02465a5a68e1
math1/computations/lrs_n6_full_triangle_span20_tail_size_two_largeB_first_row_rho200_two_parameter_tube_resume_exact_run_v26.out	204188	d9b597be0641ca31d261f2d25903acda0973d2ee6852a8c34
math1/computations/lrs_n6_full_triangle_span20_tail_size_two_largeB_first_row_rho200_boundary_verifier.py	33522	bd8e57a05f3827322d6785b63fb349be0bcc01cb54a9ef596
math1/computations/lrs_n6_full_triangle_span20_tail_size_two_largeB_first_row_rho200_boundary_exact_run_v7.out	3097	8fdbbe89d5b0b2450d665c2727a536ed17b92b2e6301b5940
math1/computations/row22_31_event263_I1_pair_resultant_4_9_bottom_after_event263_top.py	49661	76cd90ce5b2ec1a698727d2649d3e86787d3e1e37f30c747b
math1/computations/row22_31_event263_I1_pair_resultant_4_9_bottom_after_event263_top_exact_run.out	12789	3e884011e452fadcf54c4db2840b8df8a6100346682d2ff18a7
math1/computations/row22_31_event263_I1_pair_resultant_4_9_bottom_after_event263_top_artifact_validation.out	2687	84e6e02ef8b3af5f195c4b07c4e4e1f4f88abc0a7e3311839
math2/research/literature/summary.md	41719	1592ae187e24fb883e7348e9d297ed73b132d47b0646ac2d9
math2/research/solve/h3_14_boundary_cyclicity_gap.md	4120330	63dce24fa11f83a707c2b8329494eedc2f298ec464e10ec7a
math2/computations/fq3081_h313_nonboundary_atlas.sidecar.json	11473	06a3386f2bd20bc9251b1cc8f145b3bd9c519bf98d6ec5fa3
math2/computations/fq3082_h313_direct_typeii_composition.sidecar.json	7622	b6d68880515648ee69d14e00690f0bf7551b66f3d957c3dd8
math2/computations/fq3221_offline_positive_discriminant_splitting_sidecar.json	10489	762e8879401373a0baf6e302592e43021469dd1fa38d2451d
math2/computations/fq3221_offline_positive_discriminant_splitting.command.log	58	9c795770d4c211042f630e307f007819d06ae96dde315c535
math2/computations/fq3222_residual_relevance_sidecar.json	11010	7a344b005361b69cc87d318d4cec9d9e5db7746d9c3966e11
math2/computations/fq3222_residual_relevance_validation.log	45	54869dfd96289160f1cd9816afe8802687fbd4dd76a9438ce
math2/computations/fq3223_h313_saddlenode_global_landing_sidecar.json	9325	94b363b0909b6174e6ee5538c3269ea35eff240c3d6531200
math2/computations/fq3223_h313_saddlenode_global_landing.command.log	1144	f8a7857fec7ef625e4c1efc812c5ac316142f9da94a8b6339
math2/research/ROUTE_LEDGER.json	1232684	cfd094e06d09df00cdd6e10061952dcfc12d283cf809536e7

References

- [1] H. Dulac, Sur les cycles limites, *Bulletin de la Societe Mathematique de France* 51 (1923), 45–188.
- [2] J. Ecalle, Introduction aux fonctions analysables et preuve constructive de la conjecture de Dulac, Hermann, 1992.
- [3] Yu. S. Ilyashenko, Finiteness theorems for limit cycles, American Mathematical Society, 1991.
- [4] R. Roussarie and C. Rousseau, Finite cyclicity of some graphics through a nilpotent point inside quadratic systems, locally cached arXiv source [1506.07104](#).
- [5] C. Rousseau, H. Shan, and X. Zhu, Finite cyclicity of elementary graphics surrounding a focus or center in quadratic systems, locally cached arXiv source [1502.00689](#).
- [6] J. Llibre, R. Ramirez, and N. Sadovskaia, Invariant algebraic curves and limit cycles for polynomial vector fields, as audited in the Route A source cache.
- [7] J. Margalef-Bentabol and D. Peralta-Salas, Realization of prescribed limit-cycle configurations by polynomial vector fields, as cited in the project convention sources.